PRACTICAL 6

CODE :

```
import threading
import time

class Philosopher(threading.Thread):
    def __init__(self, name, left_chopstick, right_chopstick):
        threading.Thread.__init__(self)
        self.name = name
        self.left_chopstick = left_chopstick
        self.right_chopstick = right_chopstick

    def run(self):
        for _ in range(5):  # Number of times each philosopher will eat
            self.dine()

    def dine(self):
        # Philosopher is thinking
        print(f"{self.name} is thinking.")
        time.sleep(1)

        # Philosopher tries to pick up chopsticks
        with self.left_chopstick, self.right_chopstick:
            print(f"{self.name} picks up left and right chopsticks.")
            print(f"{self.name} is eating.")
            time.sleep(2)  # Time spent eating
            print(f"{self.name} is done eating.")

class Chopstick:
    def __init__(self):
```

```
        self.lock = threading.Lock()

    def __enter__(self):
        self.lock.acquire()

    def __exit__(self, exc_type, exc_val, exc_tb):
        self.lock.release()

def main():
    # Create chopsticks (5 chopsticks for 5 philosophers)
    chopsticks = [Chopstick() for _ in range(5)]

    # Create philosophers (5 philosophers)
    philosophers = [
        Philosopher(f"Philosopher {i+1}", chopsticks[i], chopsticks[(i+1)%5]) for i in range(5)
    ]

    # Start philosophers (Threads)
    for philosopher in philosophers:
        philosopher.start()

    # Wait for all philosophers to finish
    for philosopher in philosophers:
        philosopher.join()

if __name__ == "__main__":
    main()
```

Output

```
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 4 is thinking.
Philosopher 5 is thinking.
Philosopher 1 picks up left and right chopsticks.
Philosopher 1 is eating.
Philosopher 3 picks up left and right chopsticks.
Philosopher 3 is eating.
Philosopher 1 is done eating.
Philosopher 1 is thinking.
Philosopher 5 picks up left and right chopsticks.
Philosopher 5 is eating.
Philosopher 3 is done eating.
Philosopher 3 is thinking.
Philosopher 2 picks up left and right chopsticks.
Philosopher 2 is eating.
Philosopher 5 is done eating.
Philosopher 5 is thinking.
Philosopher 2 is done eating.
Philosopher 4 picks up left and right chopsticks.
Philosopher 4 is eating.
Philosopher 2 is thinking.
Philosopher 1 picks up left and right chopsticks.
Philosopher 1 is eating.
```

```
Philosopher 1 is eating.
Philosopher 4 is done eating.
Philosopher 1 is done eating.
Philosopher 3 picks up left and right chopsticks.
Philosopher 3 is eating.
Philosopher 4 is thinking.
Philosopher 5 picks up left and right chopsticks.
Philosopher 5 is eating.
Philosopher 1 is thinking.
Philosopher 5 is done eating.
Philosopher 3 is done eating.
Philosopher 3 is thinking.
Philosopher 5 is thinking.
Philosopher 4 picks up left and right chopsticks.
Philosopher 4 is eating.
Philosopher 2 picks up left and right chopsticks.
Philosopher 2 is eating.
Philosopher 2 is done eating.
Philosopher 4 is done eating.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 1 picks up left and right chopsticks.
Philosopher 3 picks up left and right chopsticks.
Philosopher 1 is eating.
Philosopher 3 is eating.
```

```
Philosopher 5 is eating.
Philosopher 1 is done eating.
Philosopher 3 is done eating.
Philosopher 1 is thinking.
Philosopher 5 picks up left and right chopsticks.
Philosopher 3 is thinking.
Philosopher 5 is eating.
Philosopher 2 picks up left and right chopsticks.
Philosopher 2 is eating.
Philosopher 5 is done eating.
Philosopher 5 is thinking.
Philosopher 4 picks up left and right chopsticks.
Philosopher 4 is eating.
Philosopher 2 is done eating.
Philosopher 2 is thinking.
Philosopher 1 picks up left and right chopsticks.
Philosopher 1 is eating.
Philosopher 4 is done eating.
Philosopher 4 is thinking.
Philosopher 1 is done eating.
Philosopher 1 is thinking.
Philosopher 3 picks up left and right chopsticks.
Philosopher 3 is eating.
Philosopher 5 picks up left and right chopsticks.
Philosopher 5 is eating.
```

```
Philosopher 5 is done eating.
Philosopher 2 picks up left and right chopsticks.
Philosopher 2 is eating.
Philosopher 5 is thinking.
Philosopher 4 picks up left and right chopsticks.
Philosopher 4 is eating.
Philosopher 3 is thinking.
Philosopher 2 is done eating.
Philosopher 2 is thinking.
Philosopher 1 picks up left and right chopsticks.
Philosopher 1 is eating.
Philosopher 4 is done eating.
Philosopher 4 is thinking.
Philosopher 3 picks up left and right chopsticks.
Philosopher 3 is eating.
Philosopher 1 is done eating.
Philosopher 3 is done eating.
Philosopher 2 picks up left and right chopsticks.
Philosopher 2 is eating.
Philosopher 5 picks up left and right chopsticks.
Philosopher 5 is eating.
Philosopher 2 is done eating.
Philosopher 5 is done eating.
Philosopher 4 picks up left and right chopsticks.
Philosopher 4 is eating.
```